

Monthly Intelligence Report

Edition 003 — The Ontario–Quebec Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

24,986

996 EHV · 6,902 HV

GRID LINES MAPPED

~12,400

~105,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 24,986 substations and ~12,400 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Canada's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Golden Substation

British Columbia, British Columbia · 69 kV

0.458

R_FINAL

Medium

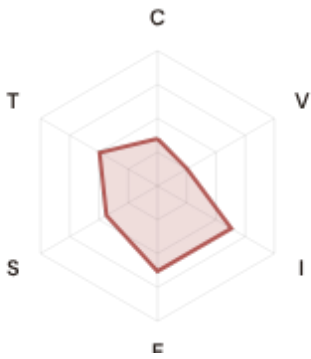
CLASSIFICATION

0.235

CI WIDTH

70th

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.348 / 0.30 (116%)

I Infrastructure: 0.632 / 0.25 (253%)

S Saturation: 0.437 / 0.20 (218%)

V Voltage: 0.253 / 0.10 (253%)

E Economic: 0.631 / 0.10 (631%)

T Transition: 0.495 / 0.05 (991%)

MODIFIERS

R3 Consequence: 0.924 ↓ dampens

R4 Graph Criticality: 1.034 ↑ amplifies

R6a Restoration: 0.982 ↓ dampens

R6b Seismic: 1.022 ↑ amplifies

R7 Digital Readiness: 1.001 → neutral

This month's spotlight — automatically selected for June 2026 — is **Golden Substation** in British Columbia, British Columbia. With an R_final of 0.458 (Medium band, 70th fleet percentile), its risk profile is dominated by the T (Transition) component at 991% of its weight ceiling, followed by E (Economic) at 631%. Modifiers collectively shift R_base by -1.8%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Canada Numérique / Canada 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1303

High/Critical + R7 < median

AVG R7 MODIFIER

1.0100

Range [0.99, 1.05]

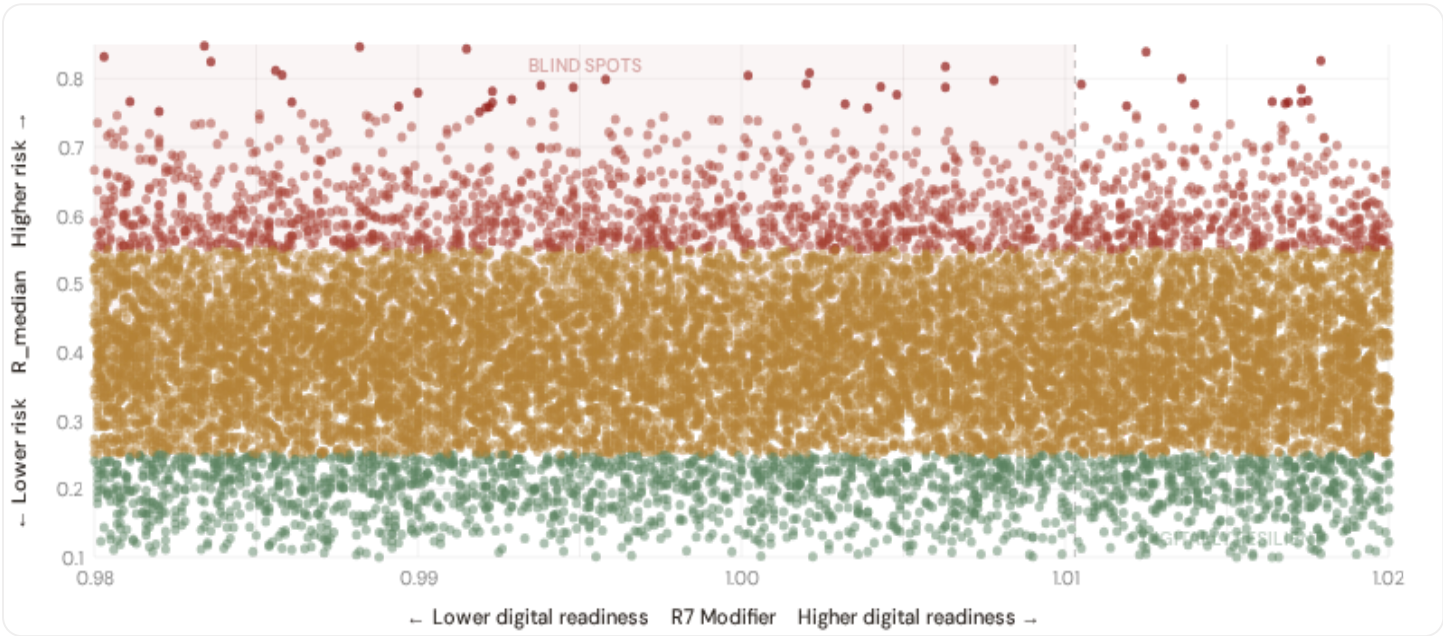
HIGH-CONSEQUENCE SUBS

6,247

25.0% · R3 ≥ 0.99

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

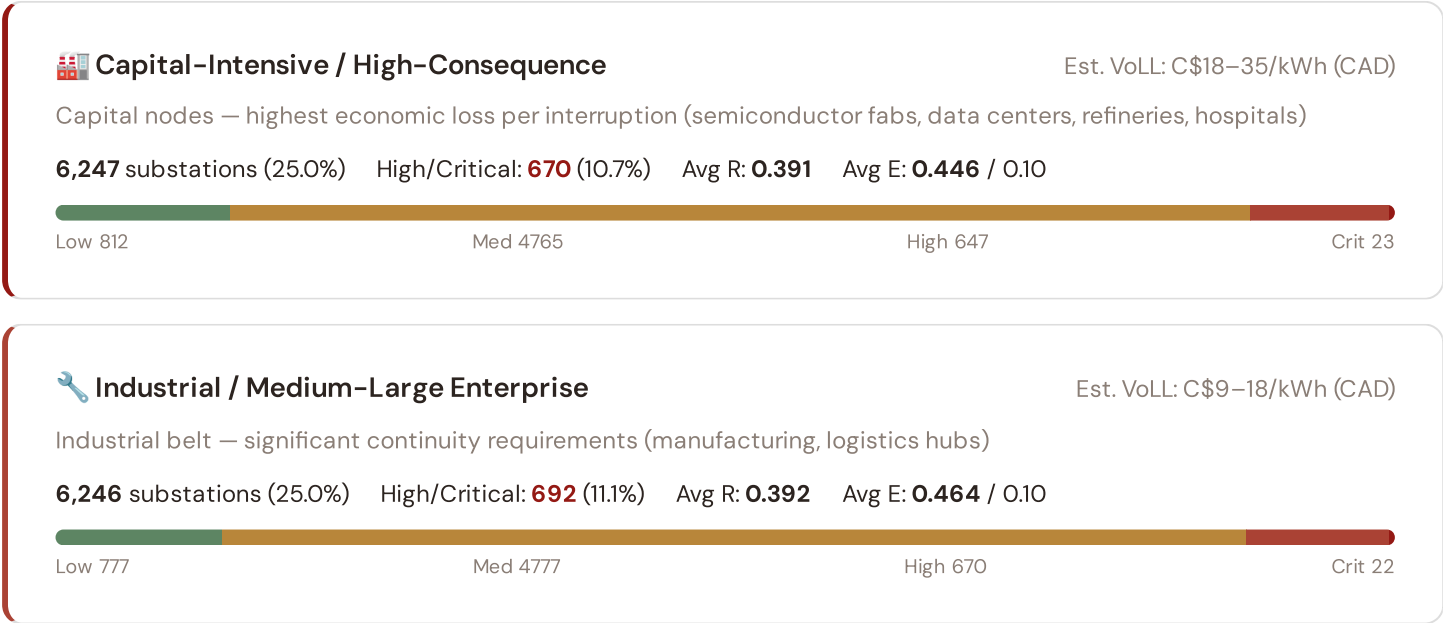


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **1303 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0103$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Ontario (412)**, **Manitoba (400)**, **British Columbia (181)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / SME-Dense

Est. VoLL: C\$4–9/kWh (CAD)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

6,248 substations (25.0%) High/Critical: **644** (10.3%) Avg R: **0.393** Avg E: **0.483** / 0.10



Light / Agricultural / Rural

Est. VoLL: C\$1–4/kWh (CAD)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

6,245 substations (25.0%) High/Critical: **640** (10.2%) Avg R: **0.389** Avg E: **0.497** / 0.10

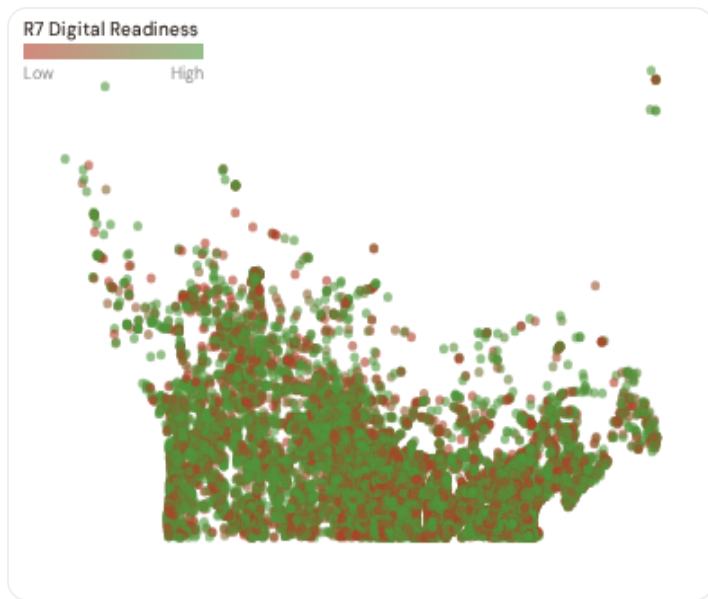


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **670 substations** combine capital-intensive economic fabric ($R3 \geq 0.99$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Ontario (2019)**, **Manitoba (1554)**, **British Columbia (1024)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Census Division Digital Readiness Ranking

Census Divisions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify census divisions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Nunavut Δ high R	1.0091
2	New Brunswick Δ high R	1.0091
3	Quebec	1.0094
4	Newfoundland and Labrador Δ high R	1.0096
5	Alberta	1.0098
6	Ontario	1.0100
7	Manitoba Δ high R	1.0101
8	Saskatchewan Δ high R	1.0102
9	Nova Scotia Δ high R	1.0105
10	British Columbia	1.0106
11	Yukon Δ high R	1.0124
12	Northwest Territories Δ high R	1.0171

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Nunavut** has the lowest average R7 (1.0091), while **Northwest Territories** leads at 1.0171 — a spread of 0.0080 across the R7 range. **5 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0100, median = 1.0103; 0 substations classified HIGH cyber-exposure; 1303 blind-spot substations (High/Critical risk + below-median R7); 33 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **20 verified public data sources** feeding 68 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

20

68 variables

WEEKLY / MONTHLY

3

High-frequency feeds

QUARTERLY / ANNUAL

10

Regular refresh cycle

STATIC / DERIVED

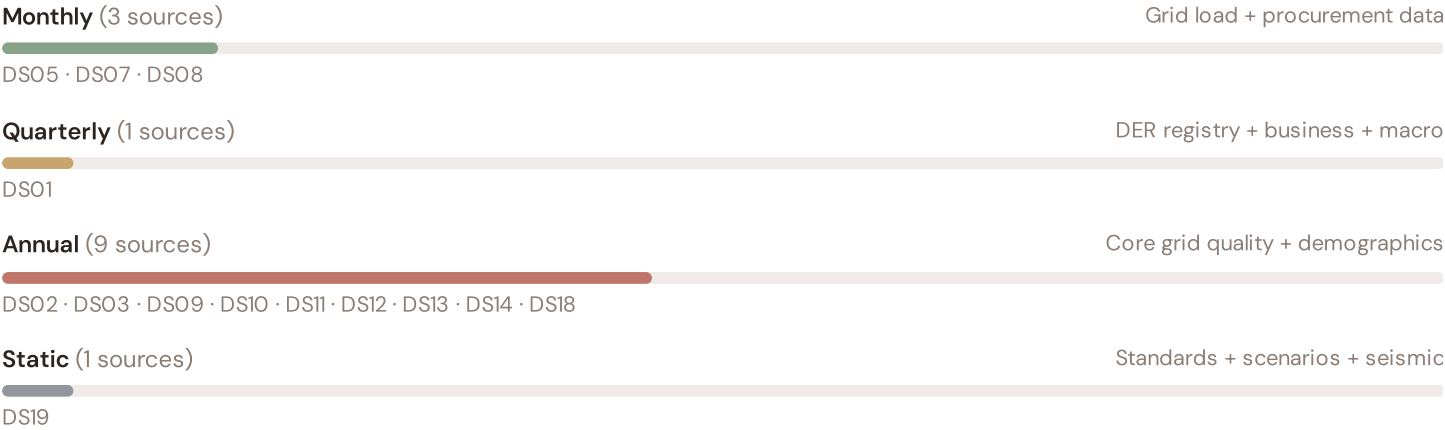
1

Data Source Registry — June 2026

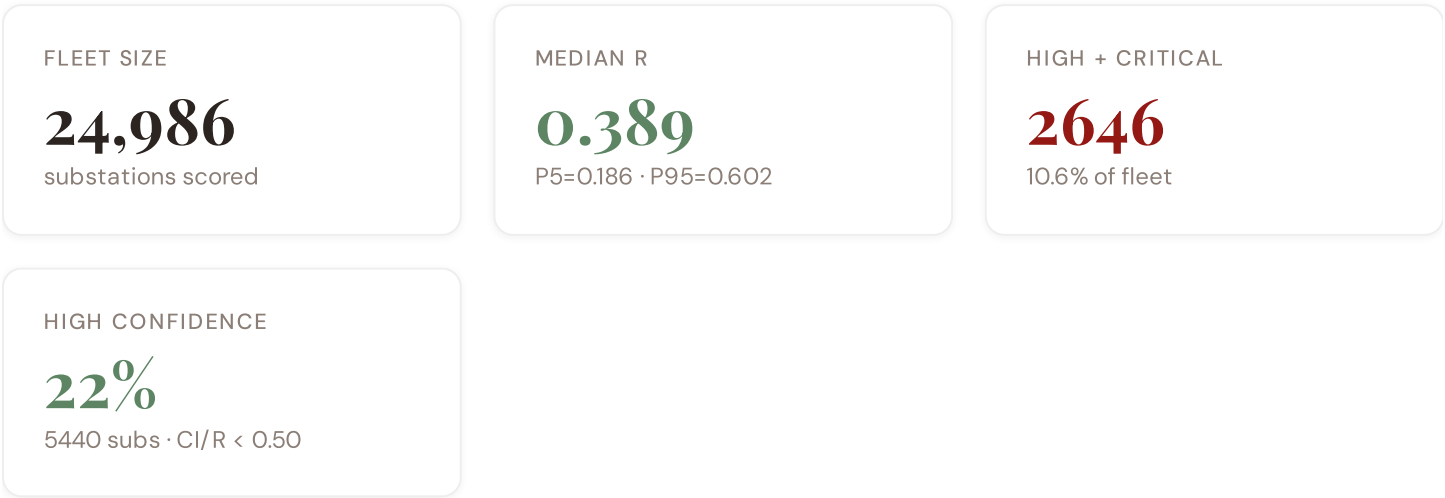
EC	Environment and Climate Change Canada (ECCC)	DAILY	Station	4 vars
HQ	Hydro-Québec (HQ)	MONTHLY	Substation	4 vars
BCH	BC Hydro	MONTHLY	Region	4 vars
AESO	Alberta Electric System Operator (AESO)	MONTHLY	Zone	4 vars
OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	3 vars
StatCan	Statistics Canada (StatCan)	QUARTERLY	Provincial	4 vars
CER	Canada Energy Regulator (CER)	ANNUAL	Utility	3 vars
NRCan	Natural Resources Canada (NRCan)	ANNUAL	National	3 vars
SKP	SaskPower	ANNUAL	Region	3 vars
MH	Manitoba Hydro	ANNUAL	Region	3 vars
NBP	NB Power (New Brunswick)	ANNUAL	Region	3 vars
NSP	Nova Scotia Power	ANNUAL	Region	3 vars
NERC	NERC (North American Electric Reliability Corporation)	ANNUAL	Regional	3 vars
IEA	IEA Global EV Data Explorer	ANNUAL	National	3 vars
CIB	Canada Infrastructure Bank (CIB)	ANNUAL	Project	3 vars
ISO9223	ISO 9223 — Corrosion Classification	STATIC	Zone	3 vars
IESO	Ontario IESO (Independent Electricity System Operator)	HOURLY	Zone	4 vars
NRCan-S	NRCan Seismic Hazard Calculator	REAL-TIME	Point	3 vars
Census	Canadian Census (StatCan)	5-YEAR	Census Division	4 vars
SSI-MC	SSI v4.0.2 Monte Carlo Engine (internal)	ON-DEMAND	Substation	4 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



C.2 Fleet Score Snapshot



BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **24,986 substation scores remain unchanged** this edition. The fleet median R of 0.389 (mean 0.391) reflects a distribution skewed toward the lower bands, with 13.1% in Low and 76.4% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across PO, P1, and P2 releases

CL01	MAJOR	Initial Canada SSI v4.0 framework release	Global
CL02	DATA	Integrated CER annual filing data for 2024	Data
CL03	DATA	Added Statistics Canada economic indicators	Data

CL04	MODEL	Calibrated component weights for Canadian grid topology	Components
CL05	DATA	Added NRCan seismic hazard zones	Infrastructure
CL06	DATA	Added Environment Canada corrosion classification	Infrastructure
CL07	DATA	Integrated flood risk from NRCan mapping	Infrastructure
CL08	DATA	Added provincial DER capacity from CER filings	Saturation
CL09	DATA	Integrated IEA EV penetration data	Saturation
CL10	MODEL	Calibrated Markov chain transition matrices for Canadian fleet	Transition
CL11	DATA	Added provincial GDP and unemployment from StatCan	Economic
CL12	MODEL	Built graph model from Canadian transmission network	Topology
CL13	QA	Monte Carlo simulation with 10k iterations	Validation
CL14	MODEL	Applied min-max scaling across Canadian substation fleet	Normalisation
CL15	DATA	Added Hydro-Quebec, IESO, BC Hydro, AESO utility feeds	Data
CL16	PATCH	v4.0.1 bug fixes and data quality improvements	Global
CL17	PATCH	v4.0.2 methodology page and dashboard alignment	Global

SECTION D — MONTHLY DEEP-DIVE

The Ontario–Quebec Corridor: Where Hydroelectric Dominance Meets Grid Modernization

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Ontario–Quebec corridor · **002** DER saturation hotspots (Alberta/Saskatchewan) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Census Division Grid Health Cards · **010** Academic validation and peer feedback · **011** North American replication feasibility · **012** Year in review.

D.1 Why the Ontario–Quebec, Why Now

ONTARIO–QUEBEC
SUBSTATIONS

13

HIGH BAND

8 + 4

92.3% of corridor

CORRIDOR MEDIAN R

0.643

Fleet median: 0.389

0 EHV · 13 HV

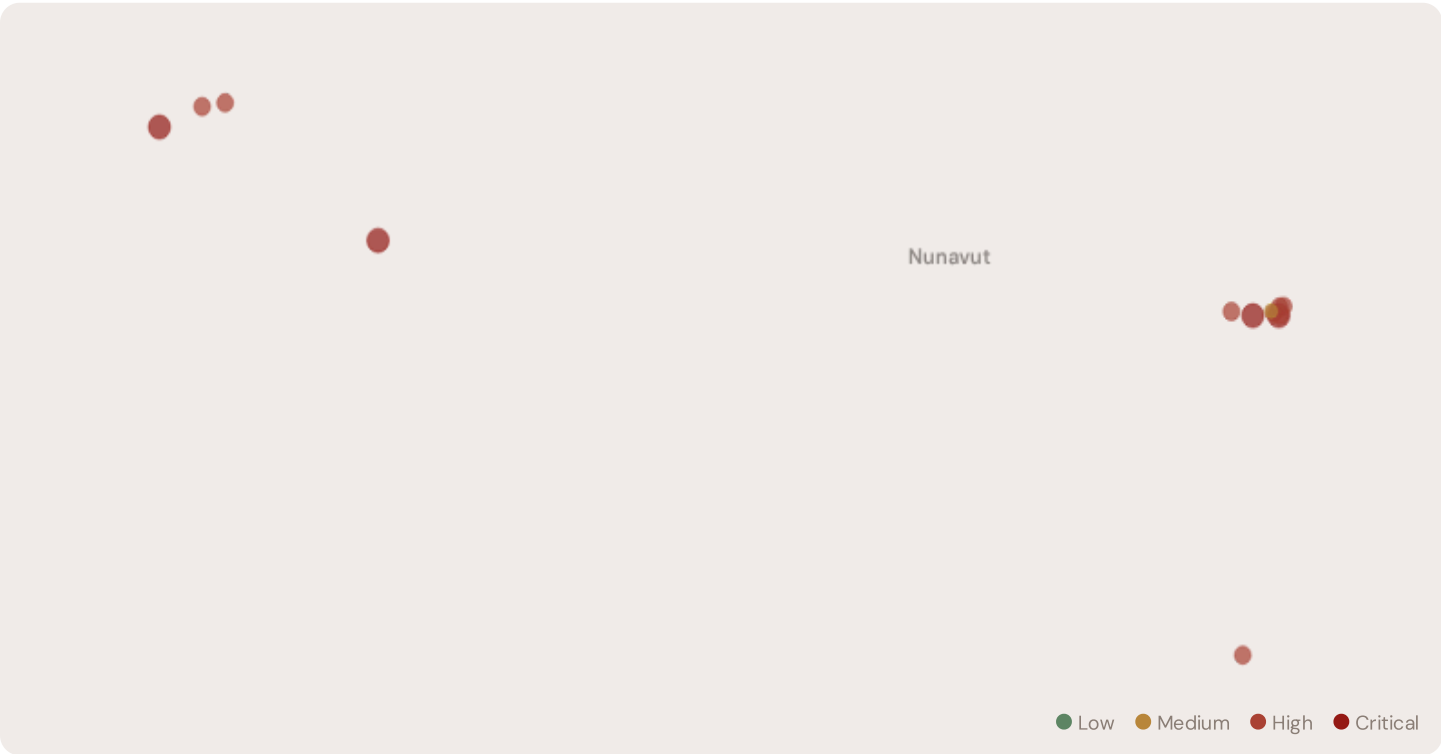
WORST CENSUS DIVISION

Nunavut

Avg R: 0.676 · 13 substations

The Nunavut hosts **13 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **8 substations (61.5%)** are classified High and **4** Critical, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.389. The corridor median R of 0.643 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Snare Forks	Nunavut	0.916	Critical	1.387	0.821	1.176	1.316	0.879	1.144	
CA_504606775	Nunavut	0.788	Critical	0.860	0.569	0.880	0.976	0.746	1.247	
CA_610310809	Nunavut	0.761	Critical	0.955	0.454	1.101	0.866	0.610	0.933	
7002S Bison Substation	Nunavut	0.752	Critical	0.825	0.647	1.182	0.845	0.808	1.382	
CA_1202723506	Nunavut	0.735	High	0.712	0.497	0.829	0.960	0.744	1.056	
CA_940055879	Nunavut	0.694	High	0.697	0.604	1.034	0.840	0.746	1.098	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
CA_1202305941	Nunavut	0.643	High	0.648	0.726	1.018	0.828	0.856	0.703	
CA_756224918	Nunavut	0.631	High	0.578	0.465	0.832	0.839	0.712	0.580	
CA_755278626	Nunavut	0.631	High	0.593	0.401	1.149	0.833	0.506	0.863	
CA_756224930	Nunavut	0.601	High	0.740	0.351	0.729	0.586	0.408	0.946	
... 3 additional substations — explore full corridor in Digital Twin →										

D.3 Component Analysis

COMPONENT AVERAGES — NUNAVUT VS FLEET

C — Continuity	0.739 vs 0.394 (+88%)
I — Infrastructure	0.956 vs 0.524 (+82%)
S — Saturation	0.675 vs 0.394 (+71%)
V — Voltage	0.536 vs 0.316 (+70%)
E — Economic	0.847 vs 0.473 (+79%)
T — Transition	0.958 vs 0.525 (+83%)

MODIFIER AVERAGES — NUNAVUT VS FLEET

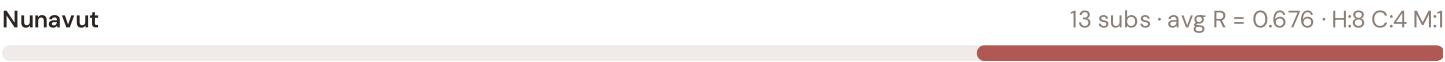
R3 Consequence	0.935	fleet 0.955 ↓
R4 Graph Crit.	0.967	fleet 0.950 ↓
R6a Restoration	0.953	fleet 0.975 ↓
R6b Seismic	1.000	fleet 1.010 →
R7 Digital Read.	1.009	fleet 1.010 →

The dominant risk driver across the Nunavut corridor is **T (Transition)**, averaging 0.958 against a fleet average of 0.525 — occupying 1917% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.847 vs fleet 0.473 (847% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.935 (fleet: 0.955), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.000, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Census Division Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The region regions-level breakdown reveals significant intra-regional variation. **Nunavut** leads with a mean R of 0.676 across 13 substations, while **Nunavut** scores 0.676 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Nunavut is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

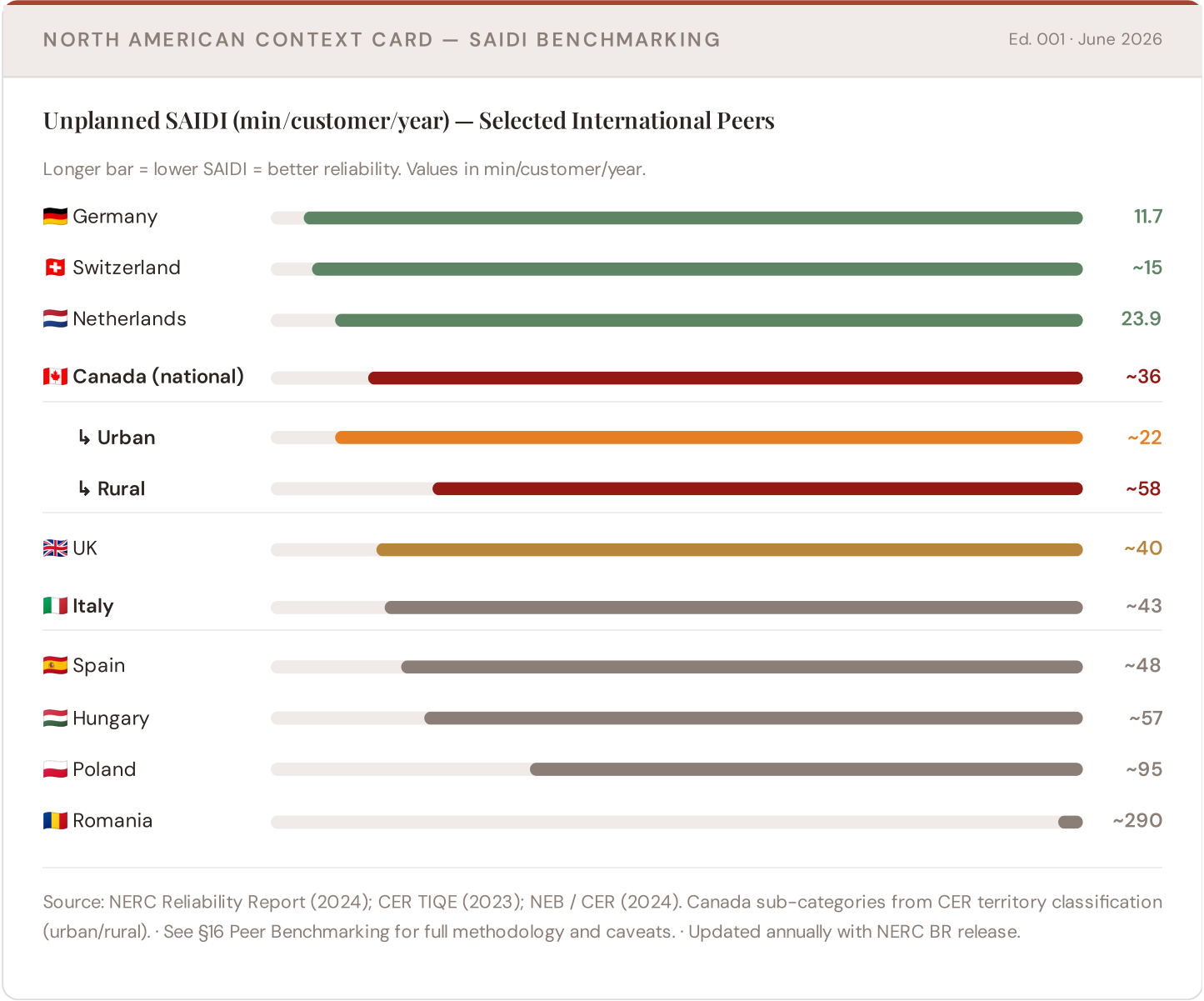
6 Nunavut substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Nunavut sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-

economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

North American Context Card

How this works: Each edition includes one North American Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Ontario–Quebec corridor analysis hinges on Canada's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs US (003), CMIP6 climate hazard vs Northern US peers (004), etc. The underlying data updates annually with the NERC Benchmarking Report release.

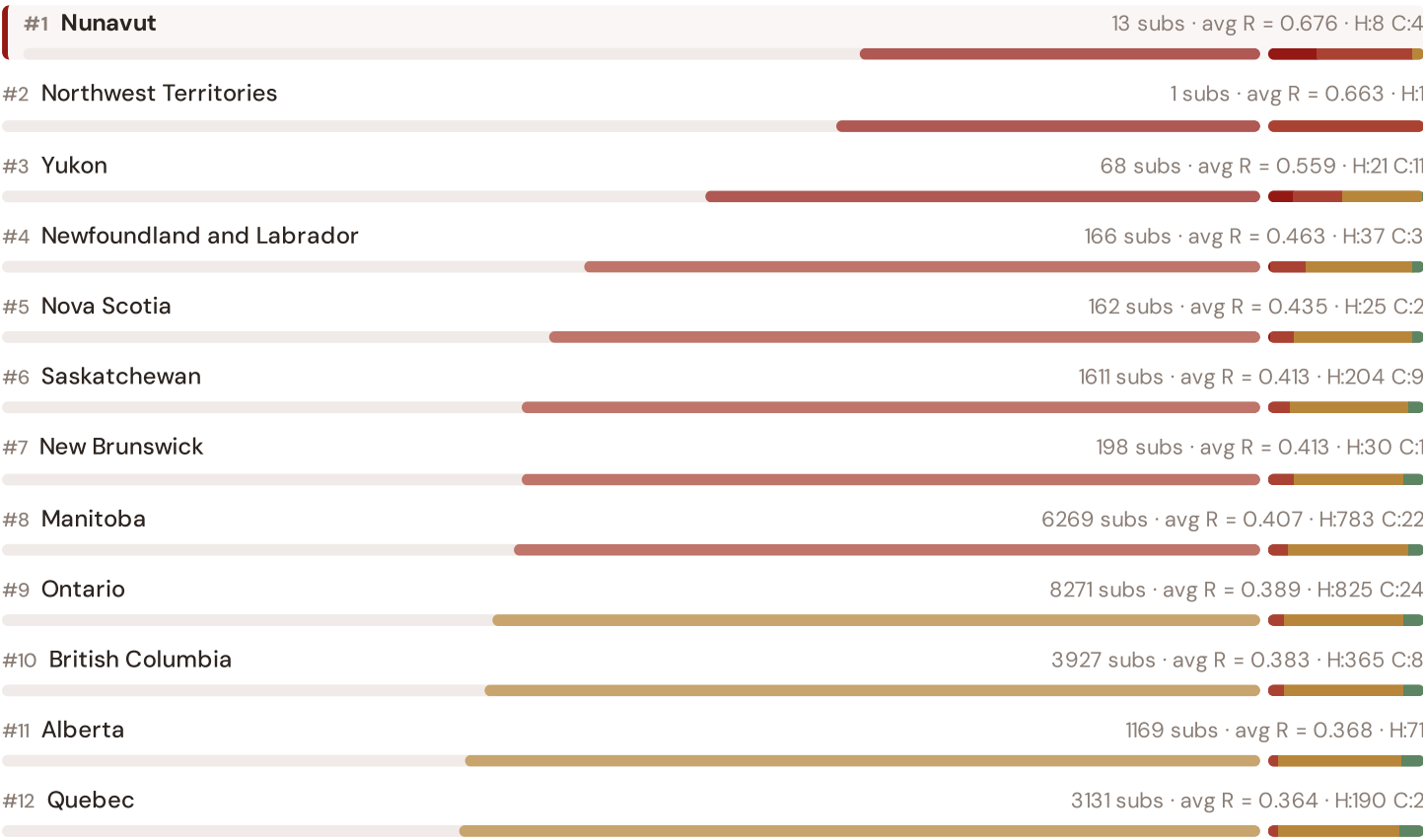


This month's key insight: The Nunavut corridor deep-dive shows **13 substations** (of 13) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.739 is **1.9× the fleet average** of 0.394. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 Canadian Provinces

Where does the Ontario–Quebec sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Nunavut ranks **#1 of 12 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Nunavut, Northwest Territories, Yukon, while the three most resilient are Quebec, Alberta, British Columbia. 8 of 12 score above the fleet average of 0.391. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Hydro One/IESO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3

§0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 24,986 substations with complete feature vectors. Average CI_width of 0.275 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 3262 Low, 19078 Medium, 2560 High, 86 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DS01 are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Canada Energy Regulator (CER)** (3 variables → C1-C4, V1-V2, I1 — outage reports, asset filings).

FLEET SPOTLIGHT

Northwest Territories: Highest Risk Concentration

Northwest Territories has the highest concentration of High/Critical substations: **1 of 1 (100%)** are in the upper risk bands — avg R = 0.663. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Canadian utility, CER, IESO, Hydro One, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

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Published 11 June 2026 · SSI Index v4.0.2 · 24,986 substations · ~12,400 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5